ROLL NO : 25CS277

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SECTION : D

SUBJECT CODE : 25CY201

SUBJECT NAME : APPLIED STATISTICS USING PYTHON

PART - B

1) Handling missing values in Pandas

Approach:

* Use dropna() to remove missing values
* Use fillna() to replace missing values.

When to use:

- dropna()

  Use when missing data is small and removing rows/columns won't significantly affect analysis.

- fillna()

  Use when data is important and should be preserved (e.g fill with mean, median, mode, or a constant).

2) Coefficient of correlation:

Given regression lines:

$$3x + 2y = 26$$
$$6x + y = 31.$$

To slope:

$$y = -\frac{3}{2}x + 13$$

Slope $b_{yx} = \frac{-3}{2}$

$$x = \frac{31 - y}{6} \Rightarrow x = \frac{-1}{6}y + \frac{31}{6}$$

slope $b_{xy} = \frac{-1}{6}$

correlation coefficient:

$$r = \pm\sqrt{b_{yx} \cdot b_{xy}} = \sqrt{\left(\frac{-3}{2}\right)\left(\frac{-1}{6}\right)}$$

$$= \sqrt{\frac{1}{4}}$$

$$r = \frac{1}{2}$$

since, both slopes are negative.

$$r = -0.5$$

3) Use of fit() and predict() in Linear regression:

fit():

* Trains the model using input data (X) and target value (Y).
* Learns coefficients (slope and intercept).

predict():

* Use the trianed model to predict output for new input data.

4) Hypothesis testing:

Given:

Total men = 600

Smokers = 325 → sample proportion $p = \frac{325}{600}$

$$p = 0.542.$$

Null hypothesis (H_0)

$H_0 : P = 0.5$ (no majority)

Alternate hypothesis (H_1)

$H_1 : P > 0.5$ (majority are smokers)

Level of significance

$\alpha = 5\%$

$Z_\alpha = 1.645$

Test statistics.

$$Z = \frac{p - P}{\sqrt{PQ/n}}$$

$$= \frac{0.542 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{600}}} = \frac{0.0417}{0.0204}$$

$Z \approx 2.06$

$Z > Z_\alpha$

$2.06 > 1.645$

H_0 is rejected, H_1 is accepted.

$\therefore$ Majority of men are smokers.

5)

Type I Error	Type II Error
• Rejecting a true null hypothesis. Example: concluding a person is guity when they are actually innocent.	• Failing to reject a false null hypothesis. Example: concluding a guitty person is innocent.

6) Measures of central tendency and dispersion.

Given:

Wages	Frequency (f)	x	fx
100 - 200	6	150	900
200 - 300	14	250	3500
300 - 400	20	350	7000
400 - 500	10	450	4500

$\Sigma f = 50$ $\Sigma fx = 15900$

i) Mean

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{15900}{50}$$

$$\boxed{\bar{x} = 318}$$

ii) Median

Wages	f	c.f
100-200	6	6
200-300	14	20
300-400	20	40
400-500	10	50

$N = 50$

$\frac{N}{2} = \frac{50}{2} = 25$

c.f just greater than $\frac{N}{2}$ is 40.

Corresponding interval is 300-400.

$$\text{Median} = l + \left(\frac{N/2 - c}{f}\right) \times i$$

$$= 300 + \left(\frac{25 - 20}{20}\right) \times 100$$

$$\boxed{\text{Median} = 325}$$

iii) Mode

Highest frequency $= 20$

$f_1 = 20$, $f_0 = 14$, $f_2 = 10$, $i = 100$, $l = 300$.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times i$$

$$= 300 + \left(\frac{20 - 14}{2(20) - 14 - 10}\right) \times 100$$

$$\boxed{\text{Mode} = 337.5}$$

Measures of dispersion:

Range

Range = L − S.

L = 500 S = 100.

Range = 500 − 100 = 400. $\boxed{\text{Range} = 400}$

Standard deviation

x	f	fx	$(x-318)$	$(x-318)^2$	$f(x-318)^2$
150	6	900	−168	28224	169344
250	14	3500	−68	4624	64736
350	20	7000	32	1024	20480
450	10	4500	132	17424	174240

$$\Sigma f(x-\bar{x})^2 = 428800.$$

$$\sigma = \sqrt{\frac{\Sigma f(x-\bar{x})^2}{\Sigma f}} = \sqrt{\frac{428800}{50}}$$

$$\boxed{\sigma = 92.6}$$

Variance.

$$\sigma^2 = (92.6)^2 = 8576.$$

$$\boxed{\sigma^2 = 8576}$$

Quartile deviation:

Q_1: $N/4 = 50/4 = 12.5$

$l_1 = 200, \quad c = 6, \quad f = 14, \quad l_2 = 300$

$$Q_1 = l_1 + \frac{\left(r\left(\frac{N}{4}\right) - c\right)}{f}(l_2 - l_1)$$

$$= 200 + \left(\frac{12.5 - 6}{14}\right) \times 100$$

$$\boxed{Q_1 = 246.4}$$

Q_3:

$$\frac{3N}{4} = \frac{3 \times 50}{4} = 37.5$$

$$Q_3 = l_1 + \left(\frac{r\left(\frac{N}{4}\right) - c}{f}\right)(l_2 - l_1)$$

$$= 300 + \frac{37.5 - 20}{20} \times 100$$

$$\boxed{Q_3 = 387.5}$$

Quartile deviation

$$QD = \frac{Q_3 - Q_1}{2} = \frac{387.5 - 246.4}{2}$$

$$\boxed{QD = 70.55}$$

Best Measure:

Median is best because it is less affected by extreme values and represent fairness in wages.

7. Python program (pandas)

```
import pandas as pd.
north = pd.read_csv("north.csv")
south = pd.read_csv("south.csv")
west = pd.read_csv("west.csv")
```

a) Merge datasets

```
df = pd.concat([north, south, west], ignore_index=True)
```

b) Convert data column

```
df['date'] = pd.to_datetime(df['date'])
df['month'] = df['date'].dt.month.
```

c) pivot table

```
pivot = pd.pivot_table(df, values = 'delivery_time',
                       index = 'region',
                       column = 'month',
                       aggfunc = 'mean')

print(pivot)
```

Approach

Step 1 : Read and Merge data

Load each csv file (north.csv, south.csv, west.csv) into separate DataFrames. Combine them into one dataset using concat().

Step 2: Process Date column

convert the data column into datetime format using to_datetime().

Extract the month from the date and store it in a new colum.

Step 3: Create Pivot table

Use pivot_table() to calculate the average delivery time.

Set:

Rows → region

Columns → month

Values → delivery_time.

8) Correlation coefficient

x	y	xy	x^2	y^2
20	300	6000	400	90000
22	320	7040	484	102400
24	340	8160	576	115600
26	360	9360	676	129600
28	380	10640	784	144400

$\Sigma x = 120$ $\Sigma y = 1700$ $\Sigma xy = 41200$ $\Sigma x^2 = 2920$ $\Sigma y^2 = 582000$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{120}{5} = 24$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{1700}{5} = 340.$$

$$\rho = \frac{\frac{1}{n}\sum xy - \bar{x}\bar{y}}{\sigma_x \cdot \sigma_y}$$

$$= \frac{\frac{1}{5}(41200) - (24)(340)}{\sqrt{\frac{2920}{5} - (24)^2} \cdot \sqrt{\frac{582000}{5} - (340)^2}}$$

$$= \frac{8240 - 8160}{\sqrt{8} \cdot \sqrt{800}}$$

$$= \frac{80}{(2.828)(28.28)} = \frac{80}{79.97}$$

$$\boxed{\rho = 1}$$

Correlation coefficient, $\rho = 1$.

Significance:

* Perfect positive correlation
* As temperature increases, electricity consumption increases proportionally.
* No randomness or derivation in data.

Conclusion:

This is a perfect and strong positive relationship between temperature and electricity consumption.

9) Python program

Given:

$X = [5, 10, 15, 20, 25]$

$Y = [7, 14, 21, 28, 35]$

i) Correlation coefficient

```
import numpy as np
X = np.array([5,10,15,20,25])
Y = np.array([7,14,21,28,35])
r = np.corrcoef(X,Y)[0,1]
print("Correlation coefficient:", r)
```

Result:

$r = 1.0$.

Inference:

Perfect positive correlation (Y increases proportionally with X).

ii) Linear Regression Model

```
from scipy.stats import linregress.
slope, intercept, r_value, p_value,
std_err = linregress(X,Y).
print("Slope:", slope)
print("Intercept:", intercept)
```

Result:

Slope, $b = 1.4$

Intercept, $a = 0$

Regression equation: $Y = 1.4X$.

iii) Plots of Data points and Regression line.

```
import matplotlib.pyplot as plt.
y_pred = slope * x + intercept
plt.scatter(x, y, label = "Data Points")
plt.plot(x, y_Pred, color='red',
label = "Regression line")
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear Regression")
plt.legend()
plt.show().
```

Result:

plot shows all points lying exactly on a straight line.

10) Chi-square test:

Observed sales (O):

20, 30, 25, 25, 35, 45, 40, 25, 63, 75

Number of weeks (n) = 10

Null hypothesis:

H_0 : sales are uniformly distributed.

Alternate hypothesis:

H_1 : sales are not uniformly distributed.

Level of significance:

5% (or) (0.05)

Test statistic:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad \text{with } (n-1) \text{ d.f.}$$

O_i	E_i	$\frac{(O_i - E_i)^2}{E_i}$
20	38.3	8.74
30	38.3	1.80
25	38.3	4.62
25	38.3	4.62
35	38.3	0.28
45	38.3	1.17
40	38.3	0.08
25	38.3	4.62
63	38.3	15.92
75	38.3	35.18

Total $\sum O_i = 383$

$$\sum \frac{(O_i - E_i)^2}{E_i} = 77.03.$$

$$E_i = \frac{\sum O_i}{n}$$

$$= \frac{383}{10}$$

$$E_i = 38.3$$

$$\boxed{\chi^2 \approx 77.03}$$

$\alpha = 5\% = 0.05$.

$n-1$ d.f $= 10-1$ d.f $= 9$ d.f.

$t_\alpha(\nu) = t_{0.05}(9)$

$= 16.92$.

$\chi^2 > \chi^2_\alpha$

$17.03 > 16.92$.

H_0 rejected, H_1 is accepted.

$\therefore$ Sales are not uniformly distributed over the 10 weeks.